

REVIEW

Heterojunction for Boosting Photo-Oxidation of Organic Contaminants

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ABSTRACT

The unrelenting existence of organic contaminants (OC) in water systems poses a severe threat to the environment and human health. Therefore, there is a need to develop sustainable and efficient water treatment technologies. In this aspect, semiconductor (SC)-based photocatalyst (PC) has emerged as a next-generation material for the photo-degradation of OC. However, its practical application is often limited by low quantum efficiency, rapid recombination rate, and lower visible-light utilization. In this context, creating heterojunctions has gained significant attention from researchers to enhance photocatalytic performance by improving charge separation and the adsorption of photons. This article summarizes the recent advances in numerous heterojunction systems such as type-I, type-II, Z-scheme, S-scheme, Schottky, and p-n junctions, with a focus on their design, structural-activity, and interfacial charge transfer mechanisms. With emphasis on the role of band alignment and surface reactive species that boost the photo-oxidation of OC. Additionally, the design of heterojunctions, photocatalytic reaction pathways, and factors influencing degradation efficiency are discussed. Finally, the current challenges and future perspectives for the rational design of next-generation heterojunction PC for the photo-degradation of OC are highlighted. Therefore, this article offers newer insights for the development of highly efficient and scalable heterojunction-based photocatalytic systems for OC.

1 | Introduction

Global Sustainable Development Goal (SDG) cannot be achieved without ensuring the availability of safe drinking water, as clean water is a fundamental right. Despite being one of the utmost imperative goals of the United Nations 2030 (UN2030) agenda, water scarcity has increased globally as populations and industrialization have grown [1–3]. Moreover, droughts, climate change, and poor management in the current period have further stressed water resources, thereby exacerbating water scarcity. The

situation is becoming more challenging due to contamination of water resources resulting from technological advancement and industrialization [4–6]. In this context, there is a need for advanced water treatment technology that significantly improves water quality, thereby easily achieving safer drinking water.

Although numerous environmental remediation technologies, mainly adsorption, membrane separation, electrochemical oxidation, and photocatalysis (PC), have been used for the removal of environmental pollutants, including heavy metal ions (HMIs),

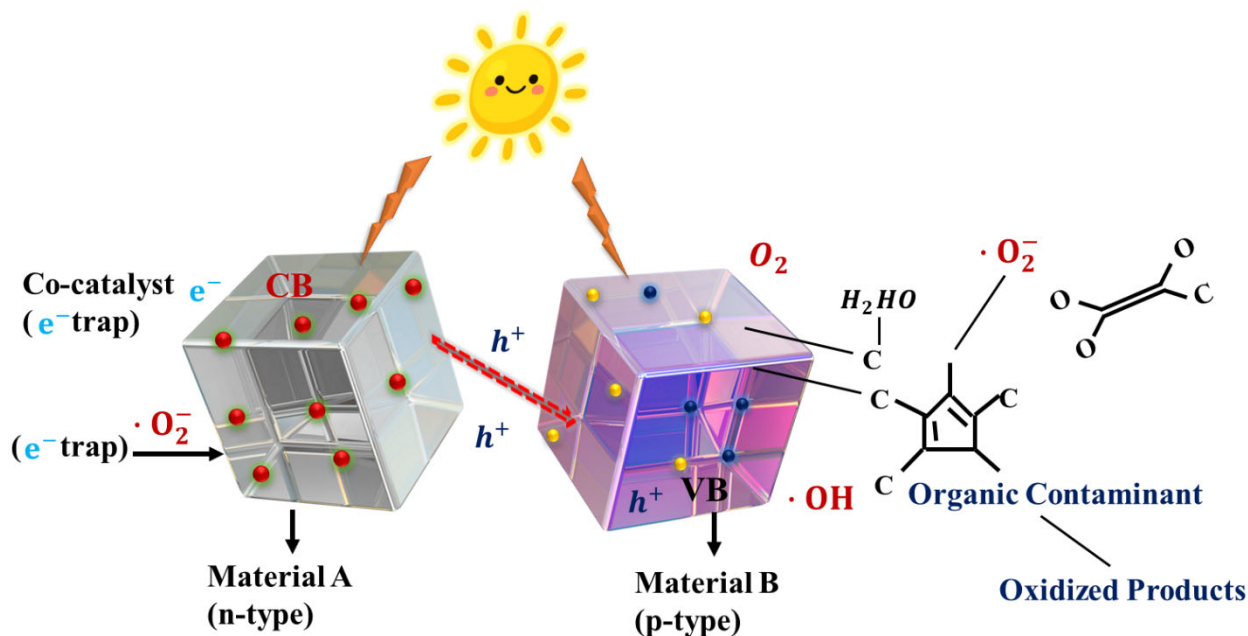


FIGURE 1 | A schematic illustration of the heterojunctions boosts the photo-oxidation of OC.

dyes, pharmaceutical compounds, and biological contaminants (BCs) [7–9]. Among all of them, the PC process has acquired substantial advantages from researchers due to its sustainability, cost-effectiveness, and rapidity. Usually, sunlight is absorbed by semiconductor (SC) materials, generating electron-hole pairs. Once the charge carrier is separated, it starts reacting with H_2O and O_2 . Holes react with H_2O molecules to form hydroxyl radical ($\cdot\text{OH}$), and electrons react with O_2 molecules to form superoxide radical ($\text{O}_2^{\cdot-}$), which degrade OC. Although the PC process is so fast, it can degrade OC within a minute or an hour. However, several disadvantages are associated with this process, such as the recombination of the generated electrons and holes, which reduces the degradation efficiency of OC. Additionally, an ideal band gap for visible-light absorption is typically in the range of 2.0–2.5 eV, so that electrons can be excited by visible radiation. Although numerous methods have been adopted to improve the degradation efficiency of OC [10–15]. Indeed, the creation of heterojunctions is an efficient method, as they substantially enhance charge separation efficiency and absorption of photons, thereby improving the degradation efficiency of OC.

Heterojunctions are interfaces between two different SC materials, where the composition changes across the interface, thereby altering energy levels and effective charge transfer. Moreover, changes in energy levels significantly reduce recombination and enhance the degradation efficiency of specific OC [16–19]. Several studies have been published so far on the formation of double/triple junctions, in which the degradation efficiency toward specific OC can be significantly increased by several folds compared to that of SC materials without a heterojunction. Numerous combinations can be used to create heterojunctions, including metal-metal, metal-polymer, and metal-ceramic junctions. Indeed, the choice of heterojunctions depends on the applications. Additionally, various combinations of OC can be used to achieve the desired properties for specific applications, such as environmental remediation, hydrogen production, CO_2

emission reduction, and solar cells [20–26]. Figure 1 shows the schematic illustration of the heterojunctions to boost the photo-oxidation of OC. This article on heterojunctions for boosting the photo-oxidation of OC is significant because it addresses the global concern of water pollution caused by persistent OC. Indeed, heterojunction PC has emerged as a promising material due to its ability to improve the absorption of light, facilitate charge separation, and suppress electron-hole recombination, thereby significantly enhancing PC oxidation efficiency. Moreover, this article provides a comprehensive framework of numerous heterojunction architectures, while elucidating their charge-transfer mechanisms and their role in producing ROS, which are responsible for the photo-degradation of OC. This review article is organized as follows: we first discuss the fundamentals of heterojunctions, heterojunction systems, and the design of heterojunctions, and their roles in the photocatalytic degradation of environmental pollutants. We then discuss how the heterojunction strategy improves the photo-degradation of OC. This article provides new insights into the types of heterojunctions and their roles in the photocatalytic degradation of OC, followed by our concluding remarks.

2 | Fundamentals of Heterojunctions in PC

2.1 | Basic Concepts of SCs

SCs are materials in which charge carriers are present, and charge flow is due to excess carriers that can participate in charge transport. Such types of charge transportation are only possible due to the presence of excessive charge carriers. The extra charge carrier is mainly due to the doping of p-type/n-type SC materials, and some impurities subsequently enhance charge transport, thereby improving performance [27–29]. In this context, charge flow might be enhanced by incorporating dopants (p-type and n-type) or by creating junctions: homo (by combining two same

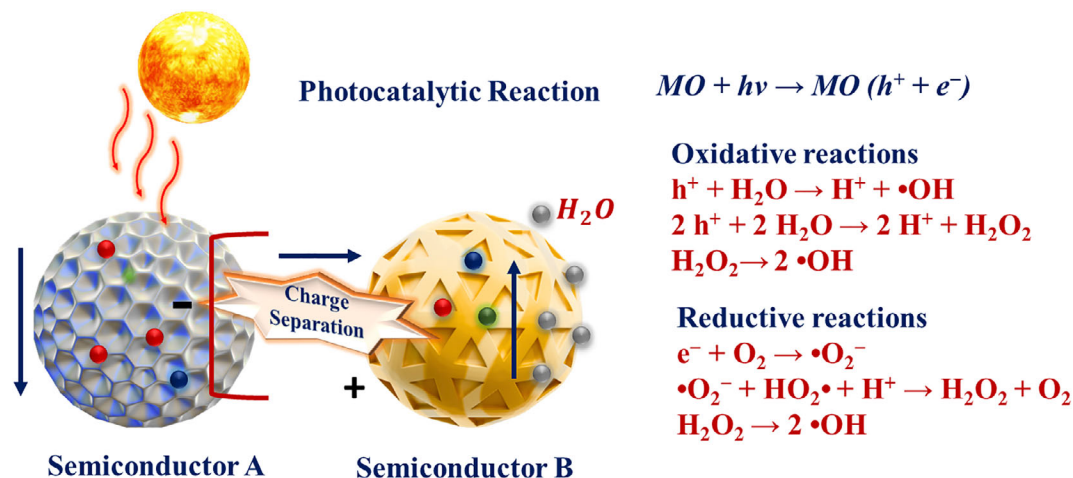


FIGURE 2 | A graphic interpretation of the basic concept of SC materials and their photocatalytic reactions.

types of dopants) and hetero (p-type and n-type). Heterojunctions are boundaries between p-type and n-type SCs that facilitate the separation of electrons and holes and are effectively used in PC. Figure 2 shows the graphical interpretation of the basic concept of SC materials and their photocatalytic reactions.

2.2 | Band Theory and Charge Transfer

Band theory describes the electronic structure of solids, particularly in crystalline materials like SCs and metals, by explaining how electrons occupy energy levels. In solids, individual atomic orbitals overlap to form continuous energy bands, which are mainly of two types. (1). Valence band (VB), and (2). Conduction band (CB). The VB is the highest range of electron energies, in which electrons are typically present at absolute zero temperature, while the CB is the lowest range of vacant electronic states [30, 31]. The energy difference between the VB and CB, or band gap, can be determined by a SC material's properties, as insulators have a large band gap value (>5 eV), thereby difficult to excite electrons, whereas SCs have a moderate band gap value (~ 0.5 – 3.0 eV), which allows electron excitation upon exposure to light/heat [32–35]. On the other hand, conductors have a ~ 0 eV band gap value, which enables the movement of electrons freely. Usually, upon photon exposure of SC materials, electrons can be excited from VB to CB, leaving behind a positively charged (hole) in VB, which is perilous for PC [36–38]. It is important to mention that narrowing the band gap value might extend absorption of the visible range, which significantly improves the performance of the PC [39–41]. For instance, SC materials with a narrowing band gap of g-C₃N₄ (~ 2.7 eV) are efficiently used in the reduction of CO₂ [42]. In general, several factors, such as material defects or band structure, influence recombination rates. SC materials are used because their band gaps enable efficient light absorption and their band-edge positions facilitate charge transfer to specific chemical species.

2.3 | Importance of Band Alignment in PC

Band alignment refers to the relative positioning of the VB and CB edges of a SC with respect to the redox potentials of the chemical

reactions involved in PC. The proper band alignment ensures that the SC materials can drive the desired redox reactions efficiently. Band alignment is vital as PC involves light-induced redox reactions, such as water splitting (H₂ production), photo-degradation of OC. For these reactions to occur, the SC's band edges must align appropriately with the redox potentials of the target reactions. The driving redox reactions are in which the CB edge must be more negative (higher in energy) than the reduction potential of the target reaction (e.g., H⁺/H₂ for hydrogen production, ~ 0 V vs. NHE at pH 0). The VB edge must be more positive (lower in energy) than the oxidation potential of the target reaction (e.g., O₂/H₂O for oxygen production, ~ 1.23 V vs. NHE at pH 0) [39, 43–45]. The aligned band gap alignment significantly enhances charge-transfer efficiency. Furthermore, misaligned bands might prevent redox reactions, rendering the materials ineffective for PC.

Such fundamentals of heterojunctions indicate that the PC performance depends on factors mainly band engineering, surface modification, morphology optimization, and interface control. Indeed, the crucial approach is picking SC materials with different types that form efficient charge pathways like type-II, Z-scheme, S-scheme, and p-n junctions. Moreover, enhancing interfacial contact by in situ growth, atomic-level bonding, and conductive layers significantly lowers the recombination rate [46–53]. Additionally, designing 1D-2D, porous, and core-shell networks considerably enhances the interfacial area and shortens the carrier diffusion distance. The doping of metals and nonmetals within the SC materials significantly improved the adsorption of the photon due to increases in the active sites and narrowing the band gap, thereby resulting in exceptional photocatalytic activity. With the help of combinational approaches, such as band alignment, strong interfacial engineering, co-catalyst deposition, defect tuning, and optimized structure, leads to efficient charge separation, reduced recombination rates, and enhanced redox activity, thereby resulting in exceptional photocatalytic performance in heterojunctions.

3 | Heterojunction Systems

A heterojunction is an interface or a boundary between two different SC materials, with varying band gaps, electronic structures,

and energy levels. Heterojunctions facilitate efficient separation of photogenerated electron-hole pairs by minimizing recombination, thereby enhancing photocatalytic activity [54]. Inspired by natural photosynthesis, two different SCs are combined to form a heterojunction with staggered band alignment, enabling efficient charge separation and strong redox potentials. Heterojunctions can also lower band gaps, which improves light absorption but may misalign band edges with redox potential, reducing reaction efficiency. Another challenge arises from surface states and defects, which can alter effective band positions and thereby impact charge transfer [55–58]. For instance, Ahmad et al. developed a TiO_2 -CdS-based heterojunction for the degradation of dye. The CdS has narrowed band gap and is susceptible to photo corrosion. The incorporation of CdS with TiO_2 shifts the photoexcited electron distribution, which significantly improves the visible light absorption ability. However, the type II charge transfer process might reduce oxidation and reduction power [59]. Stability of these heterojunctions, however, is a concern, as materials with ideal band alignment (e.g., CdS) suffer from photo-corrosion, limiting their practical use. In general, band theory explains the electronic structure of SCs, with the band gap determining their ability to absorb light and generate electron-hole pairs. The charge-transfer mechanism governs how these carriers participate in redox reactions, which are critical for PC. The efficient charge separation and minimal recombination are key to high performance. The appropriate alignment of the CB and VB edges with redox potential ensures that photocatalytic reactions can proceed [17, 60–62]. Band alignment also enhances charge separation in heterojunctions, extends light absorption, and can be tailored for specific applications, making it a cornerstone of the design of efficient PCs.

In composite PCs (e.g., TiO_2/ZnO heterojunctions), band alignment between two SCs facilitates efficient charge separation. For instance, in a Type-II heterojunction, electrons transfer from the CB of SC1 material to the CB of SC2 material with a lower energy level. The holes transfer from the VB of SC1 to the VB of SC2 with a higher energy level. This spatial separation reduces electron-hole recombination, enhancing photocatalytic efficiency. Band positions can be easily tuned through doping of nano-structure/surface modification to match specific reactions, like doping TiO_2 with nitrogen, which narrows the band gap and shifts band edges, enabling visible-light PC [63–65].

Various types of heterojunctions have been discovered, including Type I, Type II, and Type III. However, for PC, there are also two more types, including band alignment exactly equal to Type II. The mechanisms for controlling charge transfer and recombination are different. This additional type of heterojunction is the Z-scheme (which is divided into two parts, the direct and indirect ones), and another is the S-scheme (pronounced chiefly as the direct scheme, as a mediator is involved in most cases) [66]. The two new schemes are designed solely to enhance light harvesting and increase the redox ability of the designed PC.

3.1 | Designing Heterojunctions

Heterojunctions play a crucial role in the photodegradation of organic contaminants (OC) by improving charge separation and photon absorption. For instance, type II heterojunctions in

PC are more effective. Conversely, for hydrogen storage, type III modified to Z scheme heterojunctions are necessary, while type III is essential for quantum storage technologies, where the tunnelling effect is crucial [67–70]. We can notice that a particular heterojunction is typically overlooked by PC and is greatly desired for use in quantum device applications. In such a situation, how the heterojunction is designed and the charge transfer steps wholly depend upon the application requirement [71–73]. PC requires a large redox capability to either oxidize or reduce OC, and a Z-scheme heterojunction can provide such strong redox capability, which is achieved through the transfer of charges strictly in the same conduction and VBs of the SC materials. Indeed, designing the heterojunctions mainly depends on the applications, which may require competent charge transfer with strong redox potential [74–76]. Numerous studies suggested that the efficiency of heterojunctions in applications such as water splitting and photo-degradation of OC. Heterojunctions may lead to the creation of a boundary referred to as a junction. At this point, electrons and holes are brought close together to recombine, forming a depletion layer. This depletion layer creates a potential difference that inhibits other electrons from crossing and recombining at that point [77–79].

Depending on the band gap, recombination can be either radiative or nonradiative, which depends on whether the band gap is direct or indirect. To improve charge separation and extend light absorption, heterojunctions are essential for boosting the photo-degradation of OC. The interface between two SCs with different band structures allows electrons and holes to move in ways that decrease recombination and increase the production of reactive oxygen species (ROS) [80–83]. The SC with the narrower band gap in type-I heterojunctions tends to accumulate both electrons and holes, improving light absorption but frequently suffering from recombination. Type-II heterojunctions are more effective because they achieve spatial separation and enable the stronger production of radicals that attack dye molecules (like $\cdot\text{OH}$ and $\cdot\text{O}_2^-$), because electrons migrate to the material with a lower CB and holes migrate to the material with a higher VB [74, 84]. Moreover, electrons in SC1, CB recombining with holes in another SC2, VB, Z-scheme heterojunctions replicate natural photosynthesis by producing highly energetic carriers that reduce recombination and maintain strong redox ability, thereby promoting efficient photo-degradation of OC. Three SCs are combined in more intricate ternary heterojunctions to increase photocatalytic activity by expanding light absorption and establishing multiple charge transfer pathways [85, 86]. The photon absorption, charge separation, radical formation, and the subsequent photo-degradation of OC into smaller, less hazardous compounds, subsequently mineralizing them into carbon dioxide and water.

3.2 | Classification of Heterojunctions for PC

3.2.1 | Type-I (Straddling Gap) Heterojunction

Straddling Gap or type-I heterojunctions, in which VB and CB of SC1 are completely nested in SC2. The type-1 heterojunction is the simplest and is most used in PCs for the photo-degradation of OC like dyes and pharmaceutical compounds. However, confined charge carriers, poor separation, and easy recombination make

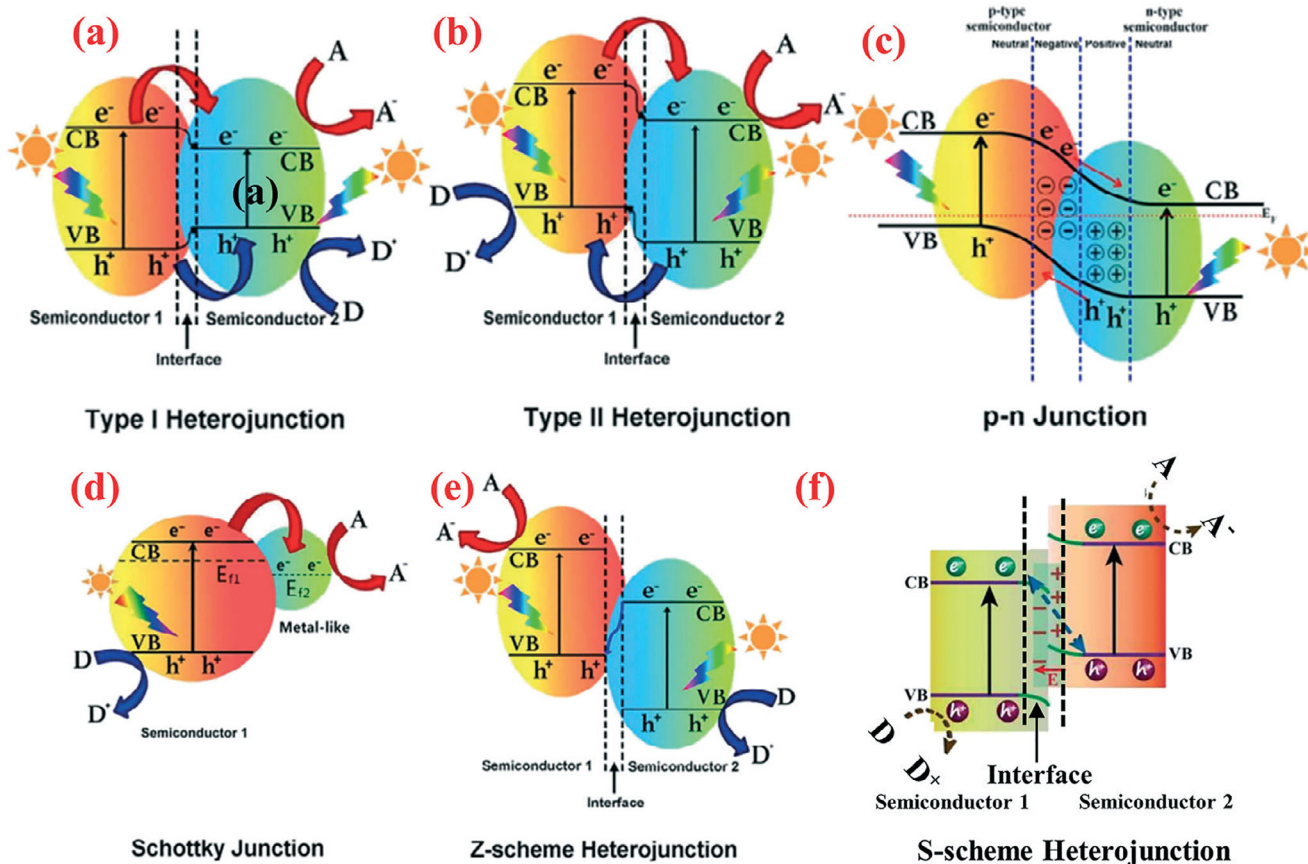


FIGURE 3 | A schematic illustration of the band structure of the different types of heterojunctions. (a) Type I, (b) Type II, (c) p-n junction, (d) Schottky junction, (e) Z-scheme, and (f) S-scheme. The image was reproduced with permission [66].

these types of heterojunctions in their least usage. Figure 3 shows the schematic illustration of the band structure of the different types of heterojunctions.

Band gap alignment determines the degradation efficiency of PC materials. Such a heterojunction follows band gap alignment, in which the VB and CB of SC1 are completely nested within the VB and CB of SC2; SC1 has a lower band gap than SC2. Flow of electrons from higher to lower energy states, that is, from CB1 to CB2, and holes prefer to flow from lower to higher energy states and will stay at VB2, which is a higher-energy state, thereby making recombination easy, with low redox potential of the charge carriers and minimal degradation of pollutants. These are most widely used in LEDs (radiative recombination), photodetectors (charge trapping), quantum dots (fluorescence in bioimaging), and passivation layers in solar cells. These heterojunctions demonstrate strong repulsion between the like-charge carriers, which somehow decreases the degradation efficiency [66]. Therefore, these types of junction-based materials are not good for the degradation of some contaminants, which need more energy to break the bonding between the molecules.

3.2.2 | Type-II (Staggered Gap) Heterojunction

Two different SCs are joined in such a way that the energy level of SC1 is higher than that of SC2. Usually, when they

meet at the junction, electrons flow from the CB of SC1 to the CB of SC2, while holes flow from the VB of SC2 to the VB of SC1 [79, 87]. Therefore, overall charge accumulation occurs at two different positions, where they cannot combine, which delays the recombination, resulting in a subsequently high photo-degradation efficiency of OC. Electrons are like water systematic down a slope (they go to the lower CB), and holes are like bubbles floating up a slope (they go to the higher VB). Type-II, the slopes guide them in opposite directions, so they don't meet again easily, and the separation makes the whole process more efficient. A separated charge carrier can now participate in different ways, as electrons can contribute to different reactions, and holes can contribute to different responses, thereby, different reaction pathways are happening together, which enhances the overall degradation efficiency toward a particular OC [18, 88, 89]. For instance, the band gap in TiO_2/CdS , and $\text{g-C}_3\text{N}_4/\text{BiVO}_4$. Such heterojunctions can be used for water-splitting applications, photo-degradation of OC, etc [90, 91]. However, inadequate redox potential reduces the charge-carrier concentration.

3.2.3 | Type-III (Broken Gap) Heterojunction

Like Type-II, but with a more extreme offset in which the CB and VB of one SC are completely misaligned with those of the other, leading to unique charge-transfer properties. This type of band gap is also known as a broken band gap, which

limits the charge-transfer behavior of type-III heterojunctions [92, 93]. Despite this, there are several advantages of these types of heterojunctions over others, such as their ability to be used in Z-scheme PC by simply incorporating a single conducting medium, mainly metallic and nonmetallic, which makes them useful in applications like PC and in quantum-based devices. Such heterojunctions are almost no charge-transfer heterojunctions with the fastest recombination rate [94–96].

Type III heterojunctions have limited applications due to their charge transfer issues. However, they can be used in modified photocatalytic systems, such as Z-scheme heterostructures, for processes like hydrogen evolution, especially when combined with a conducting medium. They also have potential in niche areas such as quantum and tunneling devices, where their unique band alignment is advantageous. However, they are not suitable for mainstream PC or solar cells, where other heterojunction types (e.g., Type I and II) are preferred [96–100].

3.2.4 | Z-Scheme

Z-scheme is a type of heterojunction PC, which mimics the natural photosynthesis “Z-scheme” (in chloroplasts) to split water or photodegrade OC using solar irradiation. Z-scheme heterojunctions are advanced photocatalytic systems inspired by the natural photosynthesis process in plants. In natural photosynthesis, two photosystems (PSI and PSII) work together to drive electron transport and convert solar energy into chemical energy [101–103]. Similarly, in artificial Z-scheme heterojunctions, two different SCs are coupled so that their band structures facilitate efficient charge separation and vigorous redox activity, mimicking the “Z”-shaped electron flow path. Usually, a Z-scheme system consists of two SCs with different band gaps that are combined. Upon light irradiation, both SCs absorb photons and generate electron-hole pairs. However, unlike conventional type-II heterojunctions, where electrons from the higher CB, SC migrate to the lower CB of the second SC, and holes do the opposite, in a Z-scheme, the photogenerated electrons in the CB of the less favorable SC recombine with holes in the VB of the more negative SC at the interface. This intentional recombination leaves behind the high-energy electrons in the CB of one SC and the strong oxidizing holes in the VB of the other, thus preserving the high redox potential of both SCs [104–108]. This mechanism enhances PC activity by retaining the strong reduction and oxidation abilities of the original SCs rather than diminishing them, as seen in traditional type-II systems. For instance, the Z-scheme composed of SC A and SC B, if A has a more negative CB and B has a more positive VB, then the recombination between A’s holes and B’s electrons leads to the accumulation of active electrons in A and active holes in B. These can then participate in strong redox reactions such as water splitting and photo-degradation of OC. There are three main types of Z-scheme heterojunctions: the traditional liquid-phase Z-scheme, the all-solid-state Z-scheme, and the direct Z-scheme. In the conventional liquid-phase system, an electron mediator like $\text{Fe}^{3+}/\text{Fe}^{2+}$ or IO_3^-/I^- is used in solution to facilitate charge transfer [109–111]. All-solid-state Z-schemes use a solid electron mediator such as a noble metal (e.g., Au, Ag) or reduced graphene oxide.

A recent development is the direct Z-scheme, which eliminates the need for a mediator and allows direct contact between the two SCs. This not only simplifies the system but also reduces electron loss and enhances long-term stability. One of the significant advantages of Z-scheme heterojunctions is their superior redox capability, due to minimal loss of charge-carrier energy. They offer enhanced visible-light absorption, longer charge-carrier lifetimes, and higher efficiency in applications such as water splitting, CO_2 reduction, hydrogen generation, and OC degradation. They are also more selective and stable than conventional heterostructures. However, there are some disadvantages, such as Z-scheme systems, especially all-solid-state and direct types, which require precise control over band alignment and intimate contact between materials [75, 112–114]. The synthesis and interface engineering are often complex, and the choice of materials must ensure appropriate band positions and chemical stability. Furthermore, in mediator-based systems, the mediators themselves can undergo side reactions or degrade over time, affecting durability. The key to a successful Z-scheme design lies in selecting SCs with suitable band edge potentials and aligning them to enable effective charge recombination and spatial separation. For example, WO_3 (with a deep VB) and $\text{g-C}_3\text{N}_4$ (with a high CB), a direct Z-scheme allows efficient separation and retention of high redox charges, ideal for PC [115, 116]. The Z-scheme concept significantly advances PC by addressing the limitations of conventional heterojunctions and enhancing solar-to-chemical energy conversion. It is a promising strategy in designing next-generation materials for sustainable environmental and energy applications.

3.2.5 | Step-Scheme (S-Scheme) Heterojunction

S-scheme heterojunctions are a comparatively newer class of heterojunction PCs developed to improve upon the drawbacks of traditional type-II and Z-scheme systems, while Z-scheme heterojunctions preserve high redox potential by mimicking the natural photosynthesis process, S-scheme heterojunctions take this a step further by introducing a more realistic energy band alignment and charge transfer mechanism based on the built-in electric field and band bending that naturally occur at SC interfaces [56, 117]. In an S-scheme system, two different SCs, typically an oxidation PC and a reduction PC, are brought into direct contact to form a heterojunction. Upon contact, due to differences in Fermi-levels, electrons from the SC with the higher Fermi level will migrate to the SC with the lower Fermi-level until equilibrium is reached, which causes band bending near the interface and establishes an internal electric field pointing from the reduction PC to the oxidation PC. When the system is exposed to light, both SCs absorb photons and generate electron-hole pairs [26, 56, 118]. However, instead of all charge carriers moving across the interface, the internal electric field and band bending cause selective recombination: photogenerated electrons in the VB of the oxidation PC recombine with holes in the CB of the reduction PC at the interface. This leaves behind high-energy electrons in the CB of the reduction PC and high-energy holes in the VB of the oxidation PC. The surviving carriers retain strong redox potential and can drive redox reactions efficiently. The term “S-scheme” refers to the S-shaped path of electrons and holes in the band diagram, where the carriers follow a step-like route from

generation to recombination to application. Unlike Z-scheme systems, which often require an external mediator or precise material contact, S-scheme heterojunctions rely on natural band bending and an electric field to direct charge transfer, making them easier to design and more robust in practice. A key advantage of the S-scheme structure is that it achieves both efficient charge separation and strong redox potential without requiring sacrificial agents or electron mediators. It also better reflects the real physical interface between two SCs, including band bending and built-in electric fields. This leads to enhanced photocatalytic activity, improved stability, and greater adaptability to different SC materials [119, 120]. The S-scheme concept allows pairing SCs that might otherwise form poor-performing type-II systems due to their reduced redox potentials. Applications of S-scheme heterojunctions include water splitting, carbon dioxide reduction, hydrogen evolution, and photo-degradation of OC. They are particularly effective under visible light, making them suitable for solar-driven environmental and energy-related technologies [121, 122]. Despite the benefits, challenges remain in accurately characterizing and proving the S-scheme mechanism in each material system. Traditional techniques such as photoluminescence, X-ray Photoelectron Spectroscopy (XPS), and Electron Paramagnetic Resonance (EPR) are used to confirm the direction of charge transfer and the presence of band bending, but detailed theoretical and experimental work is needed to validate S-scheme behavior fully. Additionally, careful selection of SCs with appropriate band positions and chemical compatibility is essential for forming an actual S-scheme interface [123]. Overall, S-scheme heterojunctions offer a more realistic and effective strategy for designing next-generation PCs, combining the best aspects of type-II and Z-scheme systems while avoiding their limitations. They represent a significant advancement in the engineering of photocatalytic materials for efficient solar energy conversion. In general, we would like to introduce the role of heterojunctions during PC.

3.3 | P-n Heterojunction

When p-type and n-type SCs come into contact, a p-n junction forms, which is a fundamental building block of most SC devices. The junction created by the joining of these two SCs possesses unique properties, which are very important for devices such as diodes, transistors, solar cells, and LEDs. Once they join, electrons start flowing from the n-side to the p-side, and the opposite pattern is followed by holes, which start flowing from the p-side to the n-side. This diffusion of the charge carriers results in the formation of a depletion region. As electrons and holes already move from here, only ionized donor and acceptor atoms remain in this depletion region, creating an internal electric field that opposes further carrier movement. This junction potential establishes equilibrium and prevents further diffusion unless an external voltage is applied [124–127]. The direction of applied voltage determines the behavior of a p-n junction. In forward bias (positive terminal to p-side, negative to n-side), the external voltage reduces the barrier potential, allowing electrons and holes to cross the junction and produce current. In reverse bias (positive to the n-side, negative to the p-side), the barrier widens, and only a minimal leakage current flows due to minority carriers. This rectifying behavior, controlling the flow of current, that is, allowing current in one direction but blocking it in the other,

makes the p-n junction function as a diode [124, 128]. In a PC, when light excites electrons in either the p or n-type material, the built-in electric field at the junction drives electrons toward the n-type side and holes toward the p-type side. This spatial separation reduces the likelihood of electron-hole recombination, allowing carriers to migrate to the surface and participate in redox reactions (e.g., oxidation by holes and reduction by electrons). The potential gradient across the junction can be tuned by material choice, improving the driving force for carrier transfer [129–132]. For example, Tamanna Gul et al. synthesized a NiO-TiO₂ composite heterojunction and applied it for the photo-degradation of the orange (II) dye. The synthesized p-n junction exhibits 96% dye degradation within 30 min under UV irradiation, which is higher than that of its bare counterpart. The higher efficiency is due to the charge-transfer, which indicates that holes are readily transported from the TiO₂ interface to the negative side (CB of NiO) via the p-n junction, while electrons are transported in the opposite direction. As a result, the electron-hole pairs are encouragingly separated by the charge gradient at the p-n junction and facilitate facile transport within the NiO/TiO₂. The electrons and holes are available for their respective photocatalytic reactions to degrade the orange dye. Herein, NiO utilized its p-type behavior, and TiO₂ acted as a typical n-type moiety [133]. Another study by Debapriya Pradhan et al. synthesized a p-n heterojunction based on TiO₂/Sr-Co₃O₄ composite for photo-degradation of MG dye under solar light irradiation. The synthesized p-n junction showed MG degradation efficiency of 92% due to the low band gap of synthesized composites (E_g~2 eV) [134]. Similarly, Junsheng Li et al. synthesized g-C₃N₄/BiOI-based p-n junction for the photo-degradation of Amaranth dye. The data suggested that the enhanced catalytic efficiency is attributed to the optimized photoelectron transfer facilitated by the p-n heterojunction formed between the g-C₃N₄ and BiOI [135]. The studies suggested that the effectiveness of the p-n junction in achieving enhanced degradation efficiency of dye-based pollutants. As p-n junctions support extending the lifetime of charge carriers and increase quantum yields, which enable better visible-light harvesting in heterojunctions. These heterojunctions also help in balancing the redox potentials, ensuring both oxidation and reduction half-reactions occur efficiently on opposite sides of the junction. Overall, p-n junctions are the most effective way to enhance performance.

3.4 | Schottky Junction (Metal-SC)

When a metal is in direct contact with an n-type SC, it forms a Schottky junction, which is a type of metal-SC junction. Schottky junctions create excellent charge transport properties due to their unique joining with metal, which is different from a p-n junction, which involves two types of SCs. This unique joining creates a difference in their work function; thereby, enhanced charge transport is observed in these types of heterojunctions. Both metal and SC, when brought into contact, electrons flow from the SC (which typically has a higher Fermi level) into the metal (lower Fermi level) until their Fermi levels align. This flow of electrons causes a depletion region to form near the interface on the SC side [136–138]. In this region, mobile electrons are depleted, leaving behind positively charged donor ions, which generate a built-in electric field and potential barrier known as the Schottky barrier. This Schottky barrier plays a central role

in determining how the junction behaves electrically. When a forward bias is applied (positive terminal to metal), the potential barrier is lowered, and electrons can move easily from the SC into the metal, allowing current to flow [139]. However, under reverse bias (negative terminal to metal), the barrier height increases, and current flow is greatly restricted. This leads to rectifying behavior, similar to a diode, but with some key differences compared to p-n junctions. One of the main advantages of Schottky junctions is their fast-switching speed. Since there are no minority carrier injections (like holes in an n-type SC), the junction doesn't suffer from charge storage effects, allowing it to switch on and off much faster than a p-n diode. This makes Schottky diodes ideal for high-frequency and high-speed electronic applications. The height of the Schottky barrier depends on the work function of the metal and the electron affinity of the SC. Metals with a higher work function form a higher barrier on n-type SCs and a lower barrier on p-type SCs. Proper selection of the metal is crucial to engineer the desired barrier height and diode characteristics. Common metals used in Schottky contacts include platinum, gold, aluminum, and chromium. However, one limitation of Schottky diodes is their relatively higher reverse leakage current compared to p-n junction diodes. Because the barrier is typically lower than that in p-n junctions, more thermally excited electrons can cross the junction under reverse bias, which can affect performance in low-power or high-temperature applications. In terms of band structure, before contact, the metal and SC have different Fermi levels. After contact, the SC's CB bends upward (in the case of an n-type SC) near the junction to align Fermi levels. This band bending forms the Schottky barrier, which electrons must overcome to move into the metal. Schottky junctions are widely used in applications such as radio-frequency mixers, power rectifiers, solar cells, photodetectors, and logic circuits. In Schottky barrier solar cells, the junction between a transparent conducting metal and an SC allows light to generate electron-hole pairs, which are then separated by the built-in field of the junction to produce photocurrent [140, 141]. The Schottky junction is a key component in high-speed, high-frequency electronics due to its fast switching, simple fabrication, and efficient rectification. Its performance depends strongly on the metal SC interface properties, making interface engineering an important part of device design. The developed junction is responsible for the high degradation efficiency. In general, studies suggested that the Schottky junction significantly enhances the absorption of light, the generation of electrons and holes, the separation of charges, redox reactions, and the subsequent formation of reaction products, which increases the degradation efficiency.

4 | 2DMs-Based Heterojunctions for the Removal of OC

Heterojunctions have been extensively developed for the photocatalytic degradation of OC. TMDs are a major class of 2DMs, with unique properties like a high surface to volume ratio, tunable band Heterojunctions can also lower band gaps, which improves light absorption but may misalign band edges with redox potential gap, and excellent electrical property are, making them an excellent candidate as a PC for photo-degradation of OC [142–144]. Usually, 2DMs already possess a low band gap, however combining them with other materials like metal, polymer or ceramics, changing them into a high efficiency heterostructure

in which heterojunction might be developed which is going to completely change the properties of the benign material when converted to composites, in this case having a composites structure in which band gap can be tunned simply by making the heterojunction is a demanding application for environmental application [143, 145].

4.1 | MoS₂-Based Heterojunctions

MoS₂-based heterojunctions are hybrid structures formed by incorporating MoS₂ with other SC materials/metals/carbon-based materials to improve the charge separation, absorption of light, and interfacial charge transfer. The unique characteristics of MoS₂, such as high surface area, extraordinary catalytic ability, and tunable band gap makes exceptional candidate for the heterojunction system [146–149]. Moreover, MoS₂ coupled with other materials to form type-II, Z-scheme, Schottky, or p-n heterojunctions that suppress electron-hole recombination and improve redox efficiency. Such heterojunctions promote efficient carrier migration and more active sites, thereby achieving exceptional performance for the removal of OC [150–152]. For instance, Gauthaman Chandrabose et al. synthesized a 2D-MoS₂-TiO₂ nanocomposite for methylene blue (MB) degradation in wastewater. Figure 4 shows the SEM, TEM, and FFT images of the MoS₂, TiO₂, and MoS₂-TiO₂. The diameter of the MoS₂ interspace layer is increased to 0.88 nm at MoS₂-TiO₂ compared to pristine MoS₂. The widening of the interlayer spacing of MoS₂ can be explained by the solvent effect. The data suggested that the synthesized composite has 10 times larger efficiency compared to the benign one [153]. Xinbo Lv et al. synthesized MoS₂-ZnIn₂S₄, having a high surface area and abundant sites for adsorption, which increases the RhB dye degradation efficiency by ~98% compare to the MoS₂-2DMs due to the creation of heterojunctions [154]. S. Sivakumar et al. synthesized a heterojunction using MoS₂-C₃N₄. The synthesized heterojunction was again used for the photo-degradation of MB dye from wastewater. The z-type MoS₂-g-C₃N₄ heterojunction has been synthesized to tune the band gap (1.2–1.9 eV) so that the synthesized material can degrade the pollutant from the wastewater using the same type of 2DMs [155]. Another study by Karthika et al. synthesized a TiO₂-MoS₂ heterojunction, which is used for the degradation of MB dye [156]. These studies suggested that MoS₂-based heterojunctions significantly improved physico-chemical characteristics compared with those of MoS₂. Moreover, interfacial contacts facilitate efficient charge separation and accelerate electron transfer, thereby suppressing the recombination rate and subsequently enhancing photocatalytic degradation. In general, MoS₂-based heterojunctions might be a promising platform for developing high-performing materials for the removal of OC.

4.2 | MXene-Based Binary or Ternary Heterojunctions

MXene, an excellent class of ceramic materials due to their incomparable characteristics like high electrical conductivity, hydrophilicity, tunable band gap, and easily modifiable surface functionality make them an excellent candidate to be used as a catalyst/co-catalyst in heterogeneous catalysis. Such properties allow for rapid charge transfer, delaying electron-hole

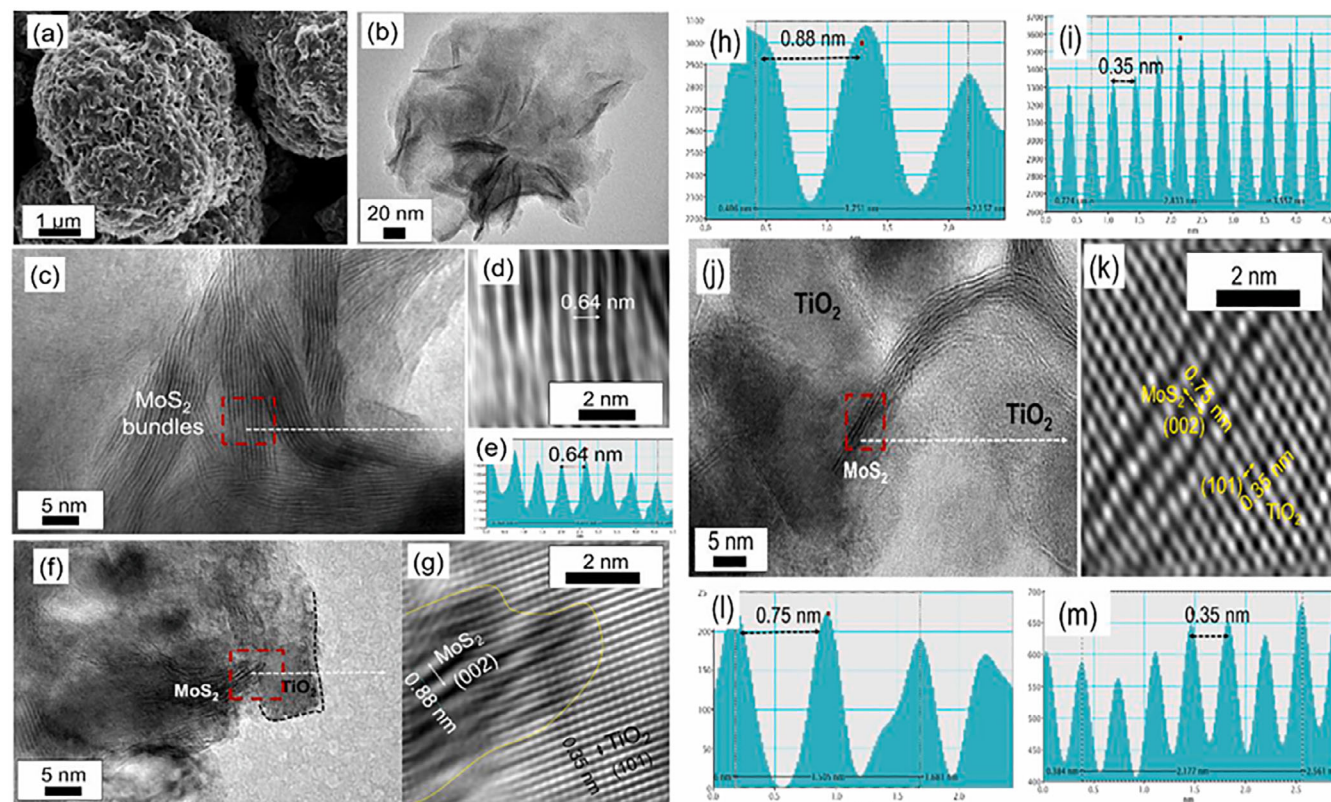


FIGURE 4 | (a) SEM image of MoS₂, (b–d) HR-TEM image of MoS₂, (e) FFT image of MoS₂, (f,g) HR-TEM image of MoS₂-TiO₂ (10%), (h) FFT image of MoS₂, (i) FFT image of TiO₂, (j,k) HR-TEM image of MoS₂-TiO₂ (50%), and (l) FFT image of MoS₂, and (m) FFT image of TiO₂. The image was reproduced with permission [153].

recombination without sacrificing redox strength in various heterojunction formations [157–159]. Indeed, literature is flooded with the use of MXene-based PCs, especially in heterojunction systems. For instance, Anand Upar et al. synthesized g-C₃N₄-CdS-MXene-based heterojunction. The synthesized heterojunction follows a Z-scheme mechanistic pathway and is applied for the MB degradation together with photoelectron catalysis (OER-ORR). The author stated that in this particular heterojunction set-up, MXene is acting like a solid-state electron mediator (Ti₃C₂Tx) [160]. Zhou et al. synthesized ZnO-NiWO₄-V₂C-MXene-based heterojunction, and used it for the degradation of cationic dyes like RhB and MB. The authors also stated that a synthesized heterojunction can facilitate efficient electron transport, improve charge separation efficiency, and provide more active sites for the photo-degradation of OC [161]. Vigneshwaran et al. synthesized (2D/2D) Ti₃C₂Tx-GO heterojunction and used it for the degradation of MB dye, due to its large surface area, exceptional hydrophilicity, different active sites, low band gap, and exceptional electrical conductivity. MXene may operate as charge-transfer intermediaries and control their capacity to capture light, while their functionality governs how they interact with other materials, in a heterojunction-based system [162]. Chen et al. synthesized g-C₃N₄-Ti₃C₂Tx-Cu₂O-based Z-scheme heterojunction. The data suggested that the Z-scheme heterojunctions preserve high-energy electrons and strongly oxidative holes, delivering superior redox capacity and durability. The synthesized junction was applied to the methyl orange (MO) dye [163]. Song et al. synthesized a β-FeOOH-N-P-MXene Schottky junction for the degradation of TC. The data suggested that

the high degradation efficiency was achieved with the synthesized heterojunction [164]. Sawaran Singh et al. developed MXene-MIL-100(Fe)-Bi₂WO₆-based heterojunction [165]. Qian et al. synthesized a g-C₃N₄-Ti₃C₂-TiO₂-based ternary junction for rhodamine B (RhB) degradation. The data suggested that the interlayer of Ti₃C₂ was inserted by the linear g-C₃N₄ to form a tightly interwoven composite interface. Figure 5 shows the schematic illustration of the proposed mechanism of the Z-scheme heterojunction structure of g-C₃N₄-Ti₃C₂-TiO₂. The Ti₃C₂ is partially oxidized to TiO₂ during the synthesis process, thereby forming a Z-scheme heterojunction structure of g-C₃N₄-Ti₃C₂-TiO₂, which acts as an electron mediator. The Z-scheme heterojunction significantly suppresses the complexation of photo-generated electrons and holes. Furthermore, formation of active species such as •O₂⁻ and •OH is produced in the system to efficiently degrade RhB dye [166]. Sahebeh Alizadeh Asli et al. synthesized Bi₂MoO₆-Ti₃C₂ MXene/NiFe₂O₄-based heterojunctions for the degradation of tetracycline (TC). The junction was synthesized using a hydrothermal process, showing 96.3% degradation efficiency, which was due to the formation of a Z-scheme heterojunction [167]. These studies suggested that, like MoS₂, MXene-based heterojunctions significantly enhance characteristics compared with those without MXene. The incorporation of MXene significantly improved the surface area, hydrophilicity, active sites, narrowed the band gap, and electrical conductivity. Additionally, MXene significantly increases the charge separation and electron transfer, which suppresses the recombination rate and subsequently enhances the photocatalytic degradation of OC.

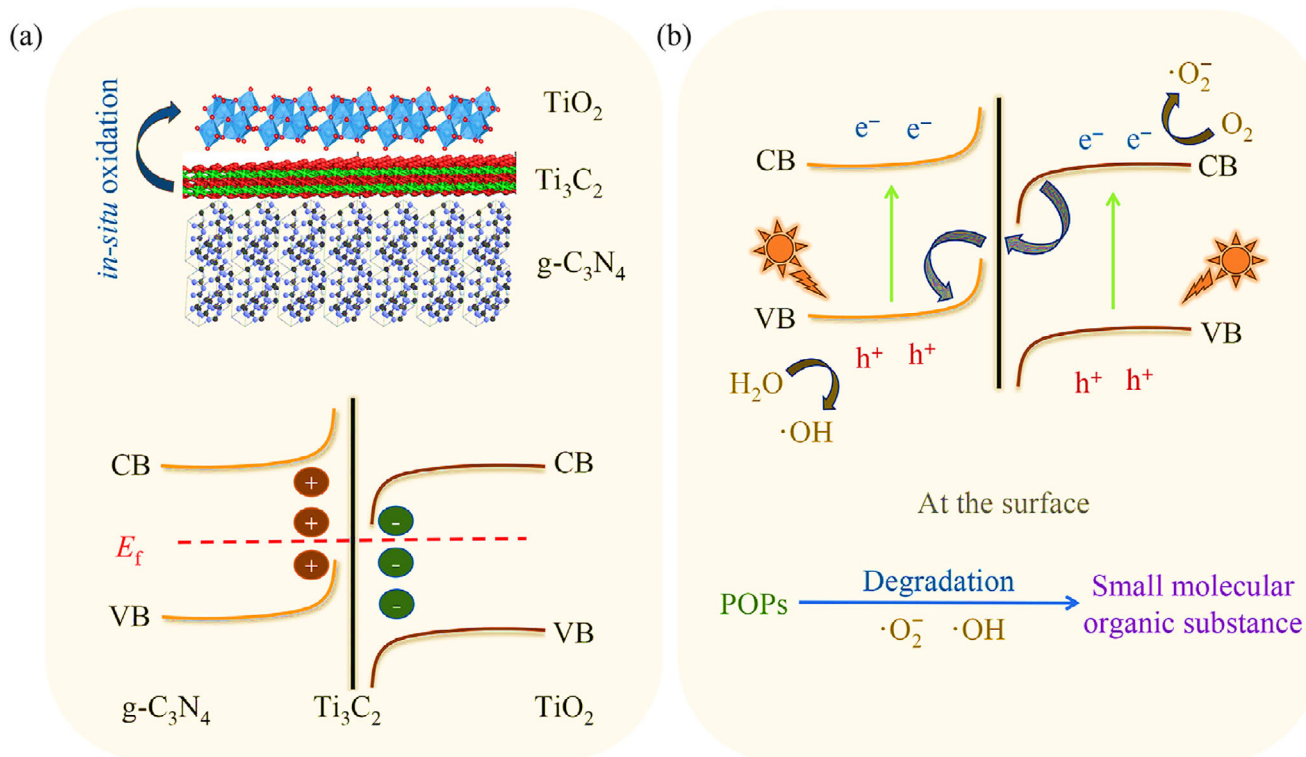


FIGURE 5 | A schematic illustration of the proposed mechanism of the Z-scheme heterojunction structure of g-C₃N₄/Ti₃C₂/TiO₂. (a) charge separation and transfer, and (b) a plausible mechanism of the Z-scheme heterojunction. The figure was reproduced with permission [166].

4.3 | C₃N₄-Based Heterojunctions for Photo-Degradation of OC

C₃N₄ is another 2DMs having a band gap of 2.6 eV, is one of the simplest known materials, and can be easily synthesized by hydrothermal synthesis. The C₃N₄ can be easily used to form heterojunctions with various types of other micro/nanomaterials and can be effectively used for the degradation of OC [168–173]. Numerous materials in combination with C₃N₄ have been used to prepare heterojunctions; for example, Wang et al. synthesized a Z-scheme Ni-C₃N₄/P-C₃N₄ heterojunction and used it for the degradation of RhB and TC. The efficiency of the heterojunction was found to be 99.1% and 86.9% for RhB and TC, respectively. The highest efficiency was found to be because of the formation of a heterojunction between Ni-doped C₃N₄ and P-doped C₃N₄, here metal acts as a bridge between the two different types of material, easing the flow of charge carriers between the junction [20]. Another study by Long Bai et al. synthesized g-C₃N₄/MoO₃-type heterojunction, which follows the S-type charge mechanism, used for the degradation of TC [174]. Niasari et al. synthesized ErVO₄/g-C₃N₄ nano-PCs for OC degradation. The data suggested that the synergetic mechanism facilitates charge carrier separation with exceptional degradation efficiency [175]. Naji et al. synthesized ZnO-Cs_{0.33}WO₃-g-C₃N₄ for the photo-degradation of RhB dye. The ternary ZnO-Cs_{0.33}WO₃-g-C₃N₄ compounds follow a Z-scheme mechanism. Figure 6 shows the plausible mechanism of the ternary ZnO-Cs_{0.33}WO₃-g-C₃N₄ nanocomposite heterojunction under solar irradiation. The ZnO-Cs_{0.33}WO₃-g-C₃N₄ compounds significantly enhance photon sorption ability, enhancing separation efficiency, and reducing recombination rate, subsequently achieving high degradation efficiency of OC

[176]. Keren Shi et al. synthesized another type of heterojunctions using 2D-2D composites, which are again used for the degradation of dye [177]. The synthesized junction has efficiently helped in the transfer of charge carriers from VB to CB, thereby being effective for the degradation of azo dye with the help of the junctions. An oxygen-rich heterojunction acts like a moiety for trapping the charge carriers, thereby delaying the recombination, which ultimately takes part in the generation of reactive radicals so that they can easily degrade the pollutant from wastewater [177].

4.4 | Triple Junctions and Their Role in the Photo-Degradation of OC

Triple junctions are not unanimously better than double junctions is still an interrogation, but they are geometrically essential in a world of three or more plates. Some scientists believe that triple junctions help in producing the stair in which charge carriers can slowly flow, and this is how recombination can be increased, and also this helps in the effective use of charge carriers. Numerous studies reported by several scientist in the effective utilization of charge carriers in triple junctions, one study reported the effective utilization of triple junction using RGO and TMDs combination in which one 2D structure is connected with the another 2D structure by using metal based 2D sheet which is acting like a link between the two different 2D material so that transfer of electron would be feasible in all aspects that why the use of one conductor was adopted for joining two different 2D based SCs which is acting like a bridge between the two different types of 2D structure in which one is the donor of

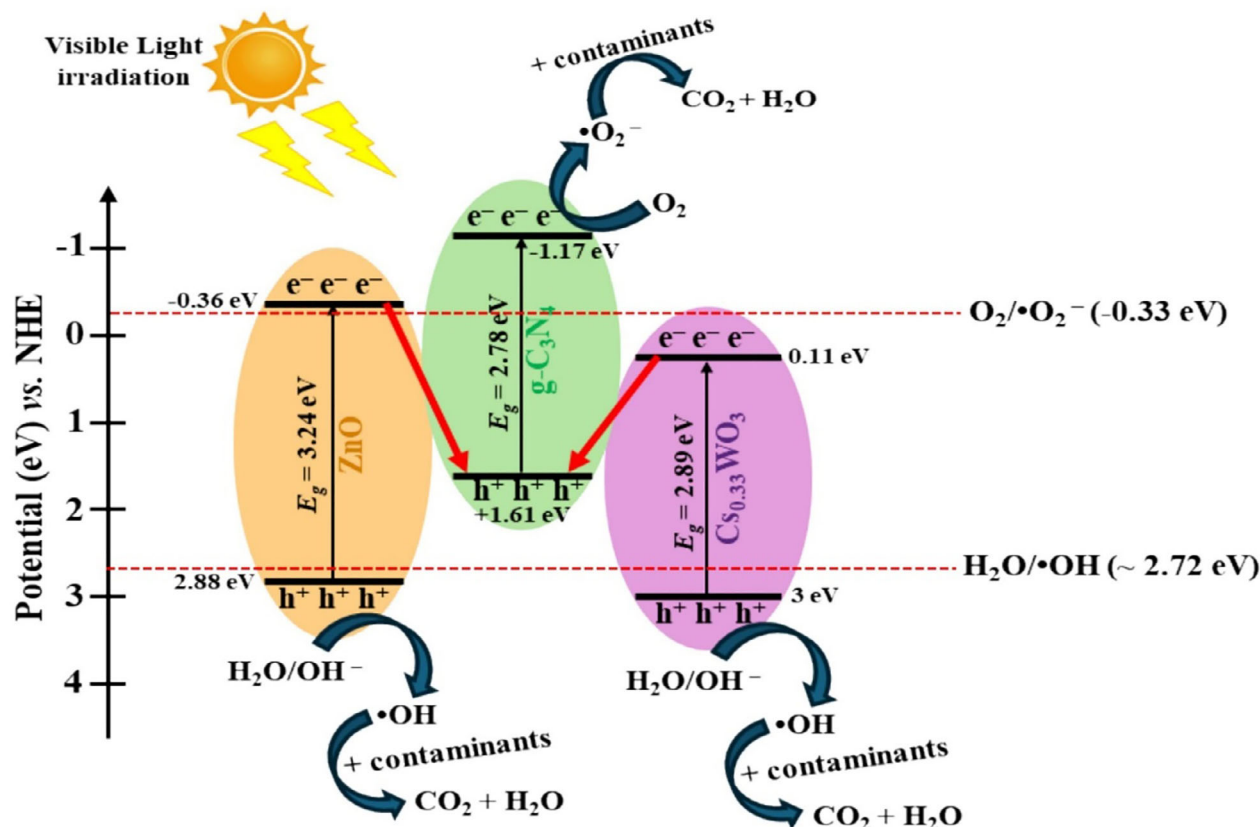


FIGURE 6 | A plausible mechanism of the ternary ZnO/Cs_{0.33}WO₃/g-C₃N₄ nanocomposite heterojunction under solar irradiation. The image was reproduced with permission [176].

electron and other one is the acceptor of electron and the link of conducting link is giving them support so that effective utilization of charge can be done with the help of one intermediate which itself is present in the form of two-dimensional (2D) layer, hence that's why it is known that triple junctions effectively work even better than the double junction where no link is there to connect two different types of 2D materials.

Several studies reported the use of ternary junctions in which the use of TMDs based ternary junctions have been reported, commonly used composite materials for ternary composites have been synthesized by combining different forms of materials like 2D/3D/1D, sometime 2D/2D/2D or 2D and 3D combination, using these combinations band gap can be easily tuned, which subsequently participate in environmental remediation technology. For instance, Drmoseh et al. synthesized a Bi₂S₃-MoS₂-TiO₂-based triple junction, the synthesized triple junction is following Z-scheme configuration and is used for the photo-degradation of MB. The TMDs provide a high surface area for the photo adsorption and subsequent charge transfer [178]. Another study by Xingang Wang et al. synthesized WSe₂/In₂S₃/ZnIn₂S₄-based TMD incorporated heterojunction, in which MoS₂, which is a well-known TMD, is being utilized for the photo-degradation of MB, author stated that the higher degradation efficiency is because of the proper contact between WSe₂, In₂S₃, and ZnIn₂S₄ in the p-n-p heterojunctions that facilitates charge transfer [179]. Karthi-gaimuthu et al. synthesized g-C₃N₄ integrated with MoS₂/BiVO₄ as a 2D/2D/1D heterojunction for the photodegradation of crystal violet (CV) and RhB. The synthesized ternary junction is also

tested for the dye-sensitized solar cell. The synthesized ternary junction followed the Z-scheme catalytic mechanism for the degradation of crystal violet and RhB [180]. Sabban et al. synthesized heterojunction using agricultural waste (TiO₂/sugarcane bagasse biochar/MoS₂ heterojunction). The synthesized junction follows Z-scheme mechanism and is used for the degradation of MB dye and also applied for energy storage applications [181]. Wang et al. synthesized AgI-IT-2H MoSe₂-Bi₄O₅Br₂ heterojunction, the synthesized heterojunction follows a Z-scheme mechanism, and was used for the degradation of RhB and TC [182]. Eroglu et al. synthesized another ternary junction based on g-C₃N₄-quantum dots-rGO-BP for the degradation of OC. This particular ternary junction behaves like a ladder for the flow of charge carriers, and polar functional groups are strongly coupled with rGO-BP at the interfaces. Moreover, this interaction resulted in a heterojunction with salient photophysical properties, and the PCs were tested in the photo-degradation of MO. The synthesized triple heterojunction supports improving the degradation efficiency of the MO dye [183]. Aloni et al. synthesized g-C₃N₄/NH₂-functionalized MIL-125(Ti) heterojunction and Schottky Ti₃C₂ for the photodegradation of diclofenac. The synthesized heterojunction is a type-II-based Schottky heterojunction. These findings highlight the superiority of ternary composites for the enhanced visible-light-driven photo-degradation of OC [184]. Shen et al. synthesized triple junctions based on g-C₃N₅/TiO₂-Ov@Ti₃C₂ ternary heterojunction, which is utilized for the degradation of acetaminophen. The role of the junction in the above-mentioned study is in the formation of an S-scheme heterojunction, which facilitates the recombination of low redox potential carriers while

TABLE 1 | Different heterojunctions and their photo-degradation activity against OCs.

S. No.	Heterojunctions	OCs	Remarks	Ref.
1.	2D-MoS ₂ -TiO ₂	MB	The increase in the interlayer spacing of MoS ₂ due to the solvent effect. The synthesized 2D-MoS ₂ -TiO ₂ composite has 10 times more efficiency compared to the control.	[153]
2.	MoS ₂ -ZnIn ₂ S ₄	RhB	The high surface area and active sites significantly increase the dye degradation efficiency by ~98% compare to the MoS ₂ -2DMs.	[154]
3.	MoS ₂ -C ₃ N ₄	MB	The z-type MoS ₂ -g-C ₃ N ₄ heterojunction has been synthesized to tune the band gap (1.2–1.9 eV) so that the synthesized material can degrade the pollutant from the wastewater.	[155]
4.	g-C ₃ N ₄ -CdS-MXene	MB	The synthesized heterojunction follows a Z-scheme mechanistic pathway and is applied for the MB degradation together with photoelectron catalysis (OER-ORR).	[160]
5.	ZnO-NiWO ₄ -V ₂ C-MXene	RhB and MB	The synthesized heterojunction can facilitate efficient electron transport, improve charge separation efficiency, and provide more active sites for the photo-degradation of OC.	[161]
6.	(2D/2D) Ti ₃ C ₂ Tx-GO	MB	The heterojunction efficiently degrades MB dye due to its large surface area, exceptional hydrophilicity, different active sites, low band gap, and exceptional electrical conductivity.	[162]
7.	g-C ₃ N ₄ -Ti ₃ C ₂ Tx-Cu ₂ O	MO	The Z-scheme heterojunctions preserve high-energy electrons and strongly oxidative holes, delivering superior redox capacity and durability.	[163]
8.	β-FeOOH-N-P-MXene	TC	The synthesized Schottky junction shows the high degradation efficiency was achieved with the synthesized heterojunction.	[164]
9.	g-C ₃ N ₄ -Ti ₃ C ₂ -TiO ₂	RhB	The Z-scheme heterojunction significantly suppresses the complexation of photo-generated electrons and holes. Furthermore, formation of active species such as •O ₂ ⁻ and •OH is produced in the system to efficiently degrade RhB dye.	[166]
10.	Bi ₂ MoO ₆ ⁻ Ti ₃ C ₂ -MXene/NiFe ₂ O ₄		The heterojunctions were synthesized using a hydrothermal process, showing 96.3% degradation efficiency, which was due to the formation of a Z-scheme heterojunction.	[167]

selectively preserving high-energy electrons and holes for redox reactions. These synergistic charge dynamics, driven by built-in electric fields and the bridging effect of Ti₃C₂, establish a continuous atomic-scale electron transfer network [185]. All these studies support the incorporation of TMDs in heterojunction formation whether it is double junction or triple, because of their

2D layered structure of 2D, it provides abundant active edge sites, large surface area, and excellent electron mobility, which together promote rapid charge transfer and suppress electron-hole recombination. Their tunable and narrow bandgaps (1-2 eV) enhance visible-light absorption and allow favorable band alignment with other SCs, enabling efficient Z-scheme or S-scheme pathways.

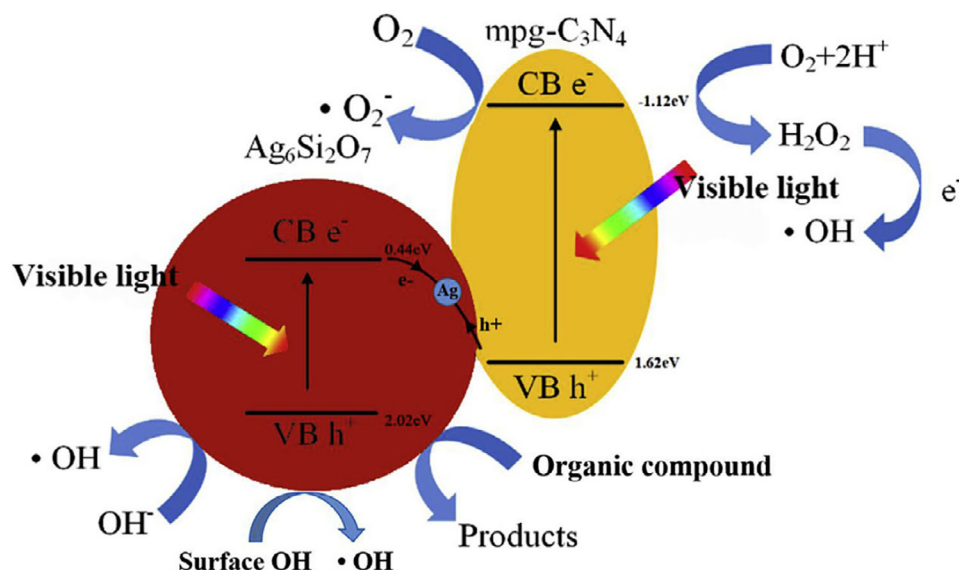


FIGURE 7 | The schematic illustration of the Z-scheme mechanism of mpg-C₃N₄/Ag₆Si₂O₇ heterojunction. The image reproduced with permission [192].

TMDs also easily form intimate van der Waals heterojunctions without lattice mismatch, ensuring strong interfacial contact in ternary systems. Additionally, their chemically stable edge sulfur/selenium sites act as catalytic centers for generating ROS like ($\bullet\text{O}_2^-$, $\bullet\text{OH}$, h^+), further accelerating photo-degradation. This combination of structural, electronic, and catalytic advantages makes TMDs one of the best materials for constructing high-performance ternary photocatalytic functions Table 1.

5 | Organic Photo-Degradation Using z-Scheme Heterojunction

Utilization of full sunlight is achieved with the use of conductive materials in designing SCs for PC applications (especially metal-based conductors), as it can reduce charge recombination. In PC, the Z-scheme heterostructure has been widely used. In this aspect, our focus is on the Z-scheme heterojunction. Numerous studies have been conducted on the Z-scheme-based heterojunction for the photo-degradation of OC, as it possesses the highest charge separation efficiency, with conducting metal bridging available for charge transport, resulting in high charge separation efficiency as well as exceptional photo-degradation efficiency [186–188]. Z-scheme follows nature's photocatalytic mechanism as it contains PS-II and PS-I, in PS-II generally available for oxidation reaction and reduction reaction occurs in PS-I. For example, Fan Wu et al. synthesized ZIF-67/BiOCl Z-Scheme heterojunction PC for the degradation of RhB and TC, in this article author achieved 97.4% RhB degradation efficiency and for TC 78.2% degradation efficiency. In this particular case, on applying electric field at the ZIF-67/BiOCl interface, photoexcited electron was transferred from BiOCl to ZIF-67, furthermore, photoexcited holes inside ZIF-67 undergo electrostatic recombination with the photoexcited electrons of BiOCl, rather than negatively charged fluorescence recombination on the surface of ZIF-67. This somehow increases the efficiency of the redox reaction and hence high degradation was achieved for RhB as well as TC [189]. Another Z scheme heterojunction based is given by Chengzhi Chu et al.

synthesized an Ag-AgBr-BiOBr-carbon nanotube (CNTs) multi-component heterojunction PC for the degradation of RhB with 95.6 % within 60 min. Interestingly, the synthesized material was able to maintain the performance up to 75.28% activity after three cycles [190]. One more example of Z-scheme photo-degradation. Taoming Yu et al. synthesized PI/TiC/TiO₂-based heterojunctions and tested them for RhB and MB dyes. The author again found high degradation efficiency due to the excellent redox ability of the synthesized heterojunction [191]. Shengnan Jin synthesized mpg-C₃N₄-Ag₆Si₂O₇ heterojunction for the photo-degradation of OC, like MB, and colorless 2,4-dichlorophenol photo-degradation [192]. Figure 7 shows the schematic illustration of the Z-scheme mechanism of mpg-C₃N₄-Ag₆Si₂O₇ heterojunctions. Yuqi Dong et al. synthesized Zn_{0.6}Cd_{0.4}S-ZnO-g-C₃N₄ dual Z-scheme heterostructures and applied them for the degradation of MB, RhB, and TC with reported efficiency of 98.52%, 99.45%, and 98.20%, for MB, RhB, and TC, respectively [193]. Baoting Tan et al. synthesized Z-scheme CN-WO₃ composites for photo-degradation of RhB with 96.5% within 2 h [194]. Mohd Amin et al. synthesized Ag-ZrO₂ heterojunction for the photo-degradation of RhB with almost 100% at 50 ppm, 10.43×10^{-5} M in 45 min. Moreover, the synthesized heterojunction effectively degrades after 5 consecutive cycles, with the degradation efficiency decreasing to 92%. The higher photo-degradation efficiency even after 5 cycles was due to the formation of a z-scheme heterojunction [195]. Zhuang et al. synthesized a CeO₂-MXene-TiO₂-based Z-scheme heterojunction for the photo-degradation of TC. The Z-scheme heterojunction mechanism enables TiO₂ CB electrons to recombine with CeO₂ VB holes, which leads to the retention of highly reductive electrons in CeO₂ and strong oxidative holes in TiO₂. The spatial charge separation prevents recombination while it increases redox capability [196].

6 | Conclusion

Heterojunction design has emerged as an overwhelming and flexible strategy to overcome the limitations of the traditional

PCs for effective photo-oxidation of OC. Indeed, constructing well-designed interfaces between two or more SCs significantly enhances absorption of photons, charge separation, and migration efficiency. Moreover, the modulation of band alignment and interfacial contact in type-II, Z-scheme, S-scheme, Schottky, and p-n heterojunctions enables the production of ROS that enhance the selective and fast photo-degradation of OC. The synthesis strategies have provided deeper insights into charge transfer pathways, which allow for high-performance and stable photocatalytic systems.

Despite such developments of SC-based PCs, numerous challenges, such as large-scale synthesis, long-term stability, photo-corrosion, interfacial recombination, and the requirement to understand mechanisms for complex pollutants. Future research should focus on the development of cost-effective and environmentally benign heterojunction materials. Moreover, integration of computational modelling with experimental studies for precise band gap engineering and development of multifunctional PC degrades the OC and recovers resources at the same time. The computation approaches, like density functional theory (DFT), are one of the most promising tools for designing heterojunctions, as DFT can predict band structures, band alignment, and charge-transfer behavior. Together, these approaches might be advantageous for designing different heterojunctions, which efficiently degrade OCs. With technological advancement in heterojunction-based photocatalyst (PC) hold great potential for practical and sustainable wastewater treatment technologies.

Author Contributions

Bhanu Sowjanya Nayudu: writing – original draft, data curation, methodology, formal analysis. **Kritika Srivastava:** data curation, writing – original draft, methodology. **Rishabh Anand Omar:** data curation, writing – original draft. **Garima Pandey:** data curation, writing – original draft, formal analysis. **Ankita Yadav:** data curation, writing – original draft. **Shruti Devi:** visualization, writing – original draft. **Mohammad Ashfaq:** conceptualization, investigation, data curation, supervision, writing – review and editing. **Werayut Srituravanich:** writing – review and editing, conceptualization. **Neetu Talreja:** conceptualization, investigation, writing – review and editing, data curation, supervision. **Divya Chauhan:** conceptualization, writing – original draft.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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